

#### 4 OLL CROSS - no yellow edge: run both



Line bar horizontal  
**F**RUR'U'F'



Hook L at front-right  
**f**RUR'U'f'

#### 5 OLL CORNERS - yellow face



Sune 1 yellow, rest CW  
**GRU**'U RU2R'



Anti-Sune 1 yellow, rest CCW  
RU2R' U'RU'R'



Pi 0 yellow, lights left  
**f**RUR'U'f' **F**RUR'U'F'



Headlights 2 yellow, lights back  
R2DR'U2 RD'R'U2R'



2x Headlights 0 yellow, lights L+R  
**GRU**'U RU'R' URU2R'



Chameleon 2 yellow adjacent  
**rUR**'U'r' **FRF**'



Bowtie 2 yellow diagonal  
**F**rUR'U'r' **FR**

#### 6 PLL CORNERS - headlights?



T headlights on ONE face - hold LEFT  
**RUR**'U' **R**'F **R2U**'R'U' **RUR**'F'



Y no headlights - diagonal swap  
**FRU**'R'U' **RUR**'F' **RUR**'U' **R**'FRF'

#### 7 PLL EDGES - put the solved edge at the BACK



Ub front edge goes LEFT  
R2U **RUR**'U' R'U' R'UR'



Ua front edge goes RIGHT  
M2U MU2 M'UM2



H opposite edges swap  
M2U'M2 U2 M2U'M2



Z adjacent edges swap  
M'U' M2U'M2U' M'U2 M2U

#### STUCK?

Corners not oriented? **Sune** until you see Sune or a solved face. · No headlights? **T** once, then look again. · No solved edge? **Ub** once, then re-read. · Nothing matches after any **U**: opposite colours facing = H, adjacent = Z.

## NOTATION

R U F L D B = turn that face clockwise · ' = anti-clockwise · 2 = half turn · M = middle slice (follows L) · x y z = turn the whole cube  
3x3: r f = 2 layers wide · big cubes: Rw = 2 layers, 3Rw = 3 layers, 2R = INNER SLICE ONLY – never widen it

1–3 BEFORE THE LAST LAYER · white on the bottom, yellow on top

white corners

RUR'U' / L'U'LU

righty / lefty – corner above its slot, repeat until it drops in

middle, to right

U RUR'U' y L'U'LU

yellow-free edge on top, colour matched to a centre

middle, to left

U' L'U'LU y' RUR'U'

mirror of the row above

Niklas

RU'L'U R'U'L

swaps a corner+edge pair, leaves the yellow face alone

– BEGINNER SHORTCUT · use it until you know steps 5–6

twist corners

R'D'RD x2

yellow sticker faces RIGHT; then U only, next corner to front-right

same, other hold

D'R'DR x2

yellow sticker faces FRONT

BIG CUBES · 4x4 & 5x5 – reduce: build 6 centres → pair the 12 edges → solve it as a 3x3

last 2 edges

Rw' U' R' U R' F R F' Rw

slice out, run it, slice back – both cubes

4x4 PLL parity

2R2 U2 2R2 Uw2 2R2 Uw2

2 edge pairs swapped · 50%

4x4 OLL parity – 1 edge pair flipped · 50%

Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

5x5 edge parity – 1 edge pair flipped · 50% · one token differs; 5x5 has no PLL parity

Rw U2 x Rw U2 Rw U2 3Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw'

Fix parity during the last layer, before OLL/PLL – both parity algs move corners.